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Original Article

ORIGINAL ARTICLE

Quantitative Radiomic Features as New Biomarkers for Alzheimer's Disease: An Amyloid PET Study

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Abstract

Growing evidence indicates that amyloid-beta (Aβ) accumulation is one of the most common neurobiological biomarkers in Alzheimer's disease (AD). The primary aim of this study was to explore whether the radiomic features of Aβ positron emission tomography (PET) images are used as predictors and provide a neurobiological foundation for AD. The radiomics features of Aβ PET imaging of each brain region of the Brainnetome Atlas were computed for classification and prediction using a support vector machine model. The results showed that the area under the receiver operating characteristic curve (AUC) was 0.93 for distinguishing AD ($N = 291$) from normal control (NC; $N = 334$). Additionally, the AUC was 0.83 for the prediction of mild cognitive impairment (MCI) converting ($N = 88$) (vs. no conversion, $N = 100$) to AD. In the MCI and AD groups, the systemic analysis demonstrated that the classification outputs were significantly associated with clinical measures (apolipoprotein E genotype, polygenic risk scores, polygenic hazard scores, cerebrospinal fluid Aβ, and Tau, cognitive ability score, the conversion time for progressive MCI subjects and cognitive changes). These findings provide evidence that the radiomic features of Aβ PET images can serve as new biomarkers for clinical applications in AD/MCI, further providing evidence for predicting whether MCI subjects will convert to AD.

Key words: Alzheimer's disease, biomarker, machine learning, prediction, quantitative radiomic features

Introduction

Alzheimer's disease (AD) is one of the most common dementia in older individuals (Handels et al. 2014). Amyloid-beta (Aβ) deposition is one of the most important neuropathological hallmarks of AD (Jack et al. 2013; Reiss et al. 2018), and the performance of classification analysis to distinguish AD and normal controls (NCs) based on positron emission tomography (PET) images have achieved an approximately 80–90% accuracy rate (Ding et al. 2019; Wang et al. 2020a). However, the application of machine learning methods in analyzing the complex patterns of PET imaging of the AD brain is in the early stages. Another concern for the application of these methods is model interoperability; these models work well for prediction, but the precise neurobiology underlying the prediction is unclear (Shen et al. 2017; Esteva et al. 2019; Kam et al. 2020; Lian et al. 2020; Tolar et al. 2020; Wang et al. 2020b; Zhu et al. 2021).

Radiomics, a method of texture analysis, refers to the extraction and analysis of a large set of advanced quantitative imaging features (Lambin et al. 2012; Gillies et al. 2016; Cook et al. 2018; Li et al. 2019c; Zhao et al. 2020). Converging evidence highlights the presence of radiomics abnormalities in AD and the possibility that textures can serve as neuroimaging biomarkers for AD (Sorensen et al. 2016; Li et al. 2019a, 2019c; Zhao et al. 2020). However, whether radiomics can be used in PET images is still debatable because of the lower resolution (Lambin et al. 2012; Cook et al. 2018; Wang et al. 2020a). The apolipoprotein E (APOE) gene is known as a major genetic risk factor for AD (Lambert et al. 2013), and the polygenic risk score (PGRS) and polygenic hazard score (PHS) provide a quantitative measure of genetic risk (Torkamani et al. 2018; Tan et al. 2019). Previous results from large-scale genome-wide association studies found associations between the PGRS and PHS for structural indices of AD brains (Axelrud et al. 2018; Tan et al. 2018; Jansen et al. 2019). However, the associations between radiomics features and genetic risk, as well as certain other clinical information, are not yet clear.

The first aim of this study was to explore whether the radiomic features of Aβ PET images can serve as neuroimaging biomarkers of AD and predictors for evaluating the progression of mild cognitive impairment (MCI) subjects. For this purpose, a classification analysis was performed, and AD subjects ($N = 291$), NCs ($N = 334$), and MCI subjects ($N = 188$) with 4 years of follow-up were included to evaluate the performance of the individual diagnosis and prediction. The second hypothesis is that the radiomic features and the classification outputs have a solid neurobiological basis, thereby showing preliminary promise as potential markers of AD or prodromal disease states. We investigated the neurobiological basis of radiomic features by relating these features to other variables, including APOE, PGRS, PHS, and other clinical measures. In particular, the classification outputs were strongly associated with the above variables and changes in the cognitive ability of the MCI subjects (Fig. 1). These comprehensive findings indicate that the radiomic features of Aβ PET images are clinically applicable neuroimaging biomarkers with a solid neurobiological foundation.

Materials and Methods

Data Acquisition and Processing

The Alzheimer's Disease Neuroimaging Initiative (ADNI) study was approved by the institutional review boards of the participating institutions. Signed informed written consent was obtained from all participants across the ADNI-1, ADNI-2, and ADNI-GO studies. In this study, 1078 (334 NC, 453 MCI, and 291 AD cases) Aβ PET images (with AV 45 tracer) were downloaded from the ADNI dataset (http://adni.loni.usc.edu). The clinical measurement data (Mini-Mental State Examination (MMSE) score, Preclinical Alzheimer Cognitive Composite (PACC) score, ADAS11, ADAS13, ADASQ4, cerebrospinal fluid (CSF) Aβ, and CSF Tau, PGRSs, and PHS) of these subjects are shown in Table 1 and Supplementary materials S1. Notably, 812 of the 1078 subjects have been previously reported (Jin et al. 2020b; Zhao et al. 2020). These prior studies focused on whether hippocampal radiomics feature (Zhao et al. 2020) or 3D attention network model (Jin et al. 2020b) based on sMRI can be biomarkers for AD, whereas the present study aimed to explore whether the radiomic features of Aβ PET images can serve as neuroimaging biomarkers of AD and predictors for evaluating the progression of MCI subjects.

The preliminary preprocessing was performed by the ADNI group (http://adni.loni.usc.edu/methods/pet-analysis-method/pet-analysis/#pet-pre-processing-container). For anatomical reference and preprocessing of the Aβ PET images, the corresponding T1 MRI image was included. The Aβ PET image was then registered to the corresponding T1 image and transformed to MNI standard space. Then, the ratio of the Aβ value was corrected for gray matter contamination using the Statistical Parametric Mapping tissue class value to reduce the partial volume effects.

Radiomic Feature Extraction from PET Images

For each subject, radiomic features from each region based on the Brainnetome atlas (246 brain regions) (Fan et al. 2016) were computed (https://github.com/YongLiulab) (Zhao et al. 2020). For each region, the 47 features (14 intensity [mainly reflecting the gray-level distribution] and 33 texture [mainly reflecting the grayscale uniformity] features) were calculated. As a result, a total of 11 562 (47×246) features were obtained for each subject. The definition of each feature can be found in previous publications (Aerts et al. 2014; Feng et al. 2018; Su et al. 2019; Zhao et al. 2020) (Supplementary materials S2). The brain region names can be found in Supplementary materials S3 (Fan et al. 2016).

Statistical Analysis—Group Difference Analysis

To verify the group differences based on radiomics features, a two-sample two-sided *t*-test of each pair of groups (NC-MCI, NC-AD, MCI-AD) was performed after regressing age and sex effects ($P < 0.05$, Bonferroni correction). We further explored whether the radiomics features were significantly different between subjects that were APOE $\varepsilon4+$ and APOE $\varepsilon4-$ in the NC, MCI, and AD groups ($P < 0.05$, uncorrected).

Classification and Prediction Analysis

To assess the multivariate performance of radiomic features and whether radiomic features can be used as biomarkers to classify NC and AD, a nonlinear support vector machine (SVM) model with a radial basis function kernel was created based on the LIBSVM library (https://www.csie.ntu.edu.tw/~cjlin/libsvm/). To compare the performance of the Aβ value and radiomics features, the same classification method used only the amyloid PET mean regional (CV2) or voxelwise SUVR intensity data (CV3) (Supplementary materials S4). In addition, a cross-validation analysis between ADNI1&GO (including 135 NCs, 221 MCI cases and 199 AD cases) and ADNI2 (including 193 NCs, 231 MCI

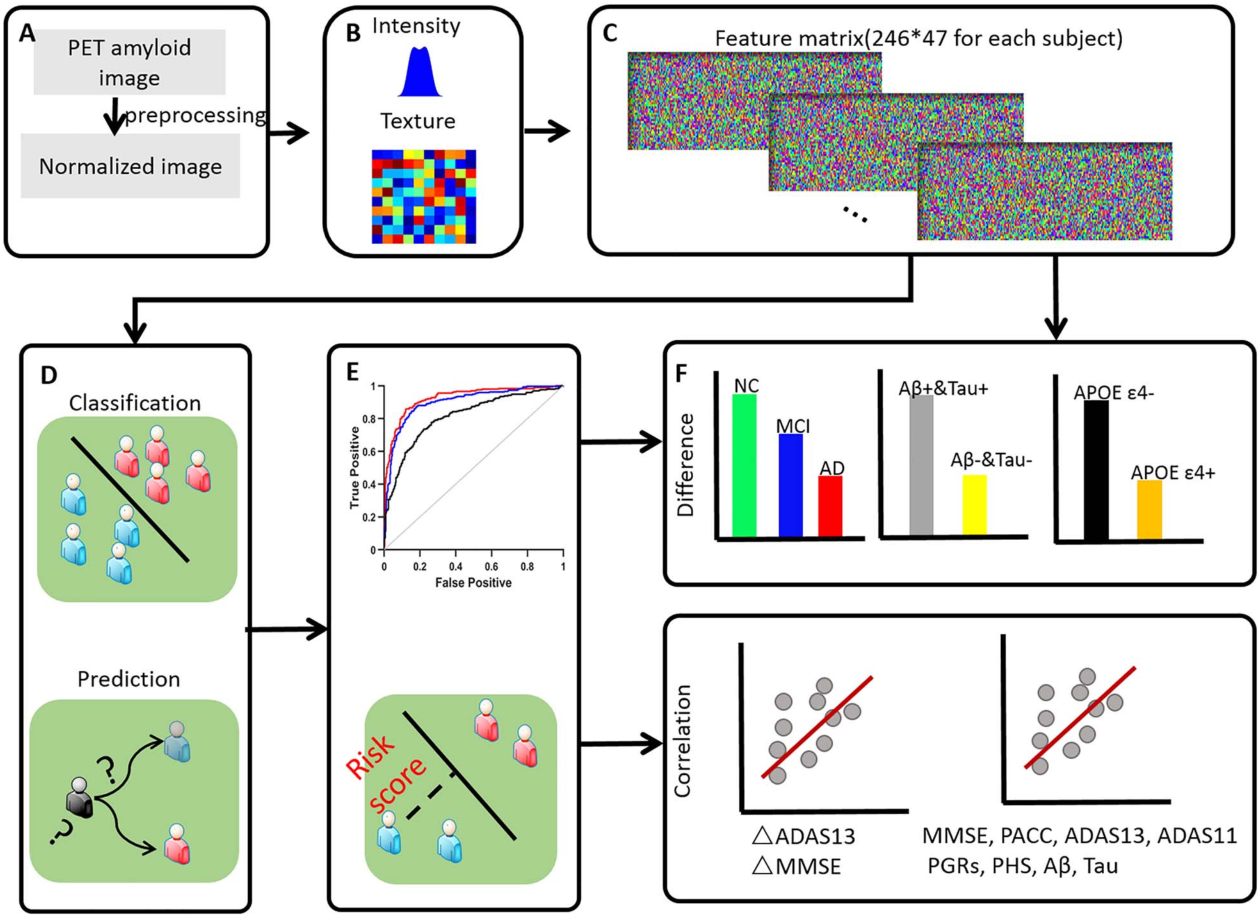

Figure 1. Schematic of the data analysis pipeline. (A) Data preprocessing, including registration and standardization with T1 image; (B) feature extraction with each brain region based on the Brainnetome atlas; (C) feature data: 246×47 matrix for each subject; (D) statistical analysis and classification analysis, including difference analysis in different groups or genetic type, correlation with MMSE or PGRS, classification of AD from NC and prediction of MCI progressing to AD in approximately 4 years.

Table 1 Detailed information regarding the subjects included in this study

		Age (years)	Sex (M/F)	MMSE
All subjects (N = 1078)	NC (334)	73.78 ± 6.01	167/167	29.10 ± 1.21
	MCI (453)	73.51 ± 6.64	272/181	27.83 ± 2.00
	AD (291)	74.69 ± 7.26	173/118	21.49 ± 4.37
	P	0.05	0.01	<0.001
Subjects with APOE $\varepsilon 4+$ (N = 477)	NC (88)	73.04 ± 6.49	41/47	29.00 ± 1.24
	MCI (195)	72.62 ± 6.22	116/79	27.46 ± 2.24
	AD (194)	74.21 ± 6.77	112/82	21.29 ± 4.57
	P	0.05	0.11	<0.001
Subjects with APOE $\varepsilon 4-$ (N = 599)	NC (245)	74.07 ± 5.83	126/119	29.14 ± 1.19
	MCI (258)	74.18 ± 6.88	156/102	28.10 ± 1.76
	AD (96)	75.68 ± 8.14	61/35	21.85 ± 3.93
	P	0.11	0.01	<0.001

cases and 92 AD cases; note that 7 subjects belonged to ADNI3, which were incorporated into ADNI2) was performed to test the robustness of the classification of AD and NC. Briefly, the classification model was created with ADNI1&GO as the training data and ADNI2 as the testing data and vice versa.

Because the number of features was approximately 10 times the sample size, and some features might not contribute to the classifier, a feature selection step was introduced by combining the *t*-test and SVM-RFE (Guyon et al. 2002). A *t*-test was performed on the training data of each feature between the AD and NC groups, and then, the features were ranked based on the T-value, with the top 2000 features selected. Next, SVM-RFE was used to rank the features, and the corresponding features were calculated in the testing data. We selected 50–250 features to create a classification model, and the accuracy (ACC), specificity (SPE), sensitivity (SEN), and area under the receiver operating characteristic (ROC) curve (AUC) were used to evaluate the performance of the classification models (Feng et al. 2018; Zhao et al. 2020). To explore whether the radiomics features can be used as biomarkers to predict whether MCI subjects will progress to AD, 88 progressive MCI (PMCI; MCI subjects who converted to AD within 4 years) and 100 stable MCI (SMCI MCI subjects who did not convert to AD within 4 years) subjects were studied. The leave-one-out cross-validation (LOOCV) (CV4) and 10-fold cross-validation with 1000 times random permutation (CV5) were introduced (Supplementary materials S4).

Relationships between Radiomic Features/Decision Value and Cognitive Ability, ADAS-Cog, Genetics, and CSF Biomarkers

To explore the relationship between identified radiomics features and clinical measures, the Pearson's correlation coefficients between the identified radiomics features and ADAS11, ADAS13, ADASQ4, CSF Aβ, and CSF Tau as well as the gene risk (PGRS and PHS) were computed ($P < 0.05$).

To verify the clinical relevance of the classification, we also investigated the correlations between the classification output and cognitive ability scores (MMSE and PACC) of the subjects ($P < 0.05$, Bonferroni corrected). Furthermore, we investigated whether there were any differences in the decision score between APOE ε4+ and APOE ε4− patients in the three groups. In addition, we explored the differences in the classification output for Aβ−&Tau−, Aβ+&Tau− (Aβ−&Tau+), and Aβ+&Tau+ subjects in the MCI group (Supplementary materials S5). To evaluate whether a disease conversion rate association could be identified, we investigated the relationship between the classification output and the length of time to convert to AD. To further study whether a disease severity association could be identified, we also investigated the relationship between the classification output and changes in cognitive ability (MMSE/ADAS13 at 12, 24, 36 months) of the MCI subjects.

Relationship between T-map of Difference Analysis and Gene Expression

Microarray data for five males and one female donor with a mean age of 42.5 years were available from the AIBS (human.brain-map.org). To explore the analogous association between the altered radiomics features in AD and gene expression, we investigated the correlations between the T-scores of these indices and the regional gene expression profiles from the AIBS (Hawrylycz et al. 2012; Whitaker et al. 2016). Gene expression was estimated by the Allen Atlas (http://human.brain-map.org/) and mapped to the Brainnetome atlas with the open tool "abagen" (https://github.com/rmarkello/abagen).

Following the study by Whitaker et al. (2016), the partial least squares (PLS) method was introduced for dimension reducing. PLS can estimate the correlation between the metrics, here, the T-map of the identified radiomics feature as an independent variable and gene expression as dependent variables.

Results

Demographic Characteristics and Neuropsychological and Aβ Assessments

As expected, the MMSE score, PACC score, PGRS, PHS, ADAS11, ADAS13, ADASQ4, CSF Aβ, and CSF Tau were significantly different ($P < 0.001$) among the NC, MCI, and AD groups (Table 1 and Supplementary materials S6).

AD patients had higher Aβ values; individuals in the NC group had lower Aβ values, and MCI patients had Aβ values in between those of the AD and NC groups. To further evaluate the altered patterns, we investigated the Aβ value differences among groups based on brain regions (Brainnetome atlas with 246 regions, $P < 0.05$ Bonferroni correction with $N = 246$). The results showed that the Aβ values in the AD group were significantly higher than those in the NC group ($P < 0.05$, Bonferroni correction with $N = 246$), especially in the precuneus area, posterior cingulate gyrus, temporal lobe, and hippocampus. Similar regions with slight alterations were found when comparing the Aβ values between the NC and MCI groups or between the MCI and AD groups (Supplementary materials S6).

Altered Radiomics Features in AD

Among all 11 562 defined features, 2659 (approximately 23%) were significantly altered among the groups ($P < 0.05$, Bonferroni corrected). When comparing the AD and NC groups, the results showed that the most significantly altered regions were the temporal lobe, the parietal lobe, and the hippocampus, with more than 23 significantly altered radiomic features (Fig. 2A and D and Supplementary materials S7). When evaluating the differences between the NC group and the MCI group (Fig. 2B and E and Supplementary materials S7) and the MCI group and the AD group (Fig 2C and F and Supplementary materials S7), we also found significant but weaker alterations than those between the NC and AD groups (10–15 features) in the abovementioned regions. The results showed that several features (such as run length nonuniformity, high gray-level run emphasis, short-run high gray-level emphasis, and long-run high gray-level emphasis) were significantly different in more than 50 brain regions between the different groups (Fig. 2D–F).

The Associations between Radiomics Features and the MMSE Score, PGRS, and APOE Gene Type

The results showed that 70% of the identified features (P value $< 0.05/11\,562$) significantly correlated with MMSE ($P < 0.05$, Bonferroni corrected), and the most significant regions that showed significant correlations with the MMSE and PACC scores were the temporal lobe, subcortical nuclei and hippocampus (more than 15 features in each brain region) (Fig. 3A and B and Supplementary materials S8).

More importantly, the regions in the middle frontal gyrus, precuneus, temporal lobe, subcortical nuclei, and hippocampus showed significant correlations with clinical measures, including ADAS11, ADAS13, ADASQ4, PGRS, PHS, CSF Aβ, and CSF Tau (more than 15 features in each brain region, $P < 0.05$, not corrected, Fig. 3C–I, Supplementary materials S8). Furthermore, the radiomics features were significantly regulated by the APOE gene, especially in the MCI group (Fig. 3J–L).

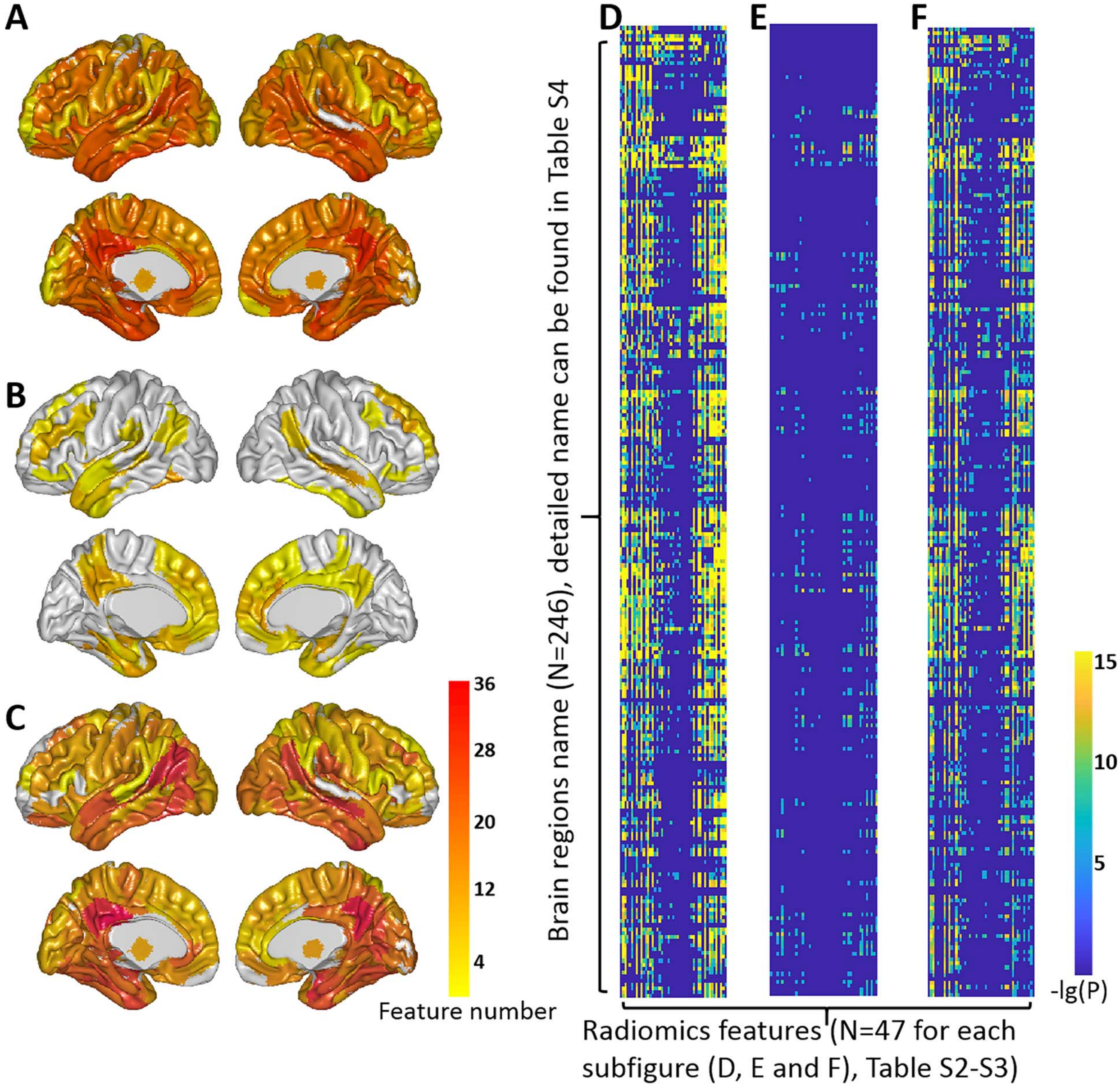

Figure 2. Radiomics features in NC, MCI, and AD. The differences in AD compared with NC (A), MCI compared with NC (B), and AD compared with MCI (C). The value of the color bar indicates the number of identified different features in each ROI in the Brainnetome Atlas. The –lg(p) of each feature in AD compared with NC (D), MCI compared with NC (E), and AD compared with MCI (F).

The Classification and Prediction Performance

For the identification of AD from NCs with the radiomics features of Aβ PET images, we obtained an ACC of 0.87 (SPE = 0.88, SEN = 0.86, AUC = 0.93) even with the standard machine learning SVM method (Fig. 4A). The performance of the classifier was evaluated using 10-fold cross-validation (CV1) with fixed hyperparameters ($c = 2$ and $g = 2^{-8}$). In fact, there was a small variance with different hyperparameters; see Supplementary material S4. In addition, we obtained an ACC of 0.89 (SPE = 0.96, SEN = 0.72, AUC = 0.94) with the ADNI1&GO as the training data and ADNI2 as the testing data. The ACC = 0.81 (SPE = 0.87, SEN = 0.72, AUC = 0.89) with the ADNI2 as the training data and ADNI1&GO as the testing data (Fig. 4C). We found that the radiomics features of the amygdala and hippocampus had a significant contribution to the classification of AD and NC, as did, in part, the inferior temporal gyrus and the parahippocampal gyrus. For the top contributing features, textural features such as "Long/Short-Run High Gray-Level Emphasis", "Gray-Level Nonuniformity", "High Gray-Level Run Emphasis","Run Length Nonuniformity" and "Maximum Probability" (details can be found in Supplementary materials S3) played an important role in the classification of AD and NC and might be caused by the abnormal deposition of Aβ.

The classification performance was better than that with only the Aβ scores based on regions (ACC = 0.75, SPE = 0.80, SEN = 0.70, AUC = 0.81) or based on voxels (ACC = 0.84, SPE = 0.86,

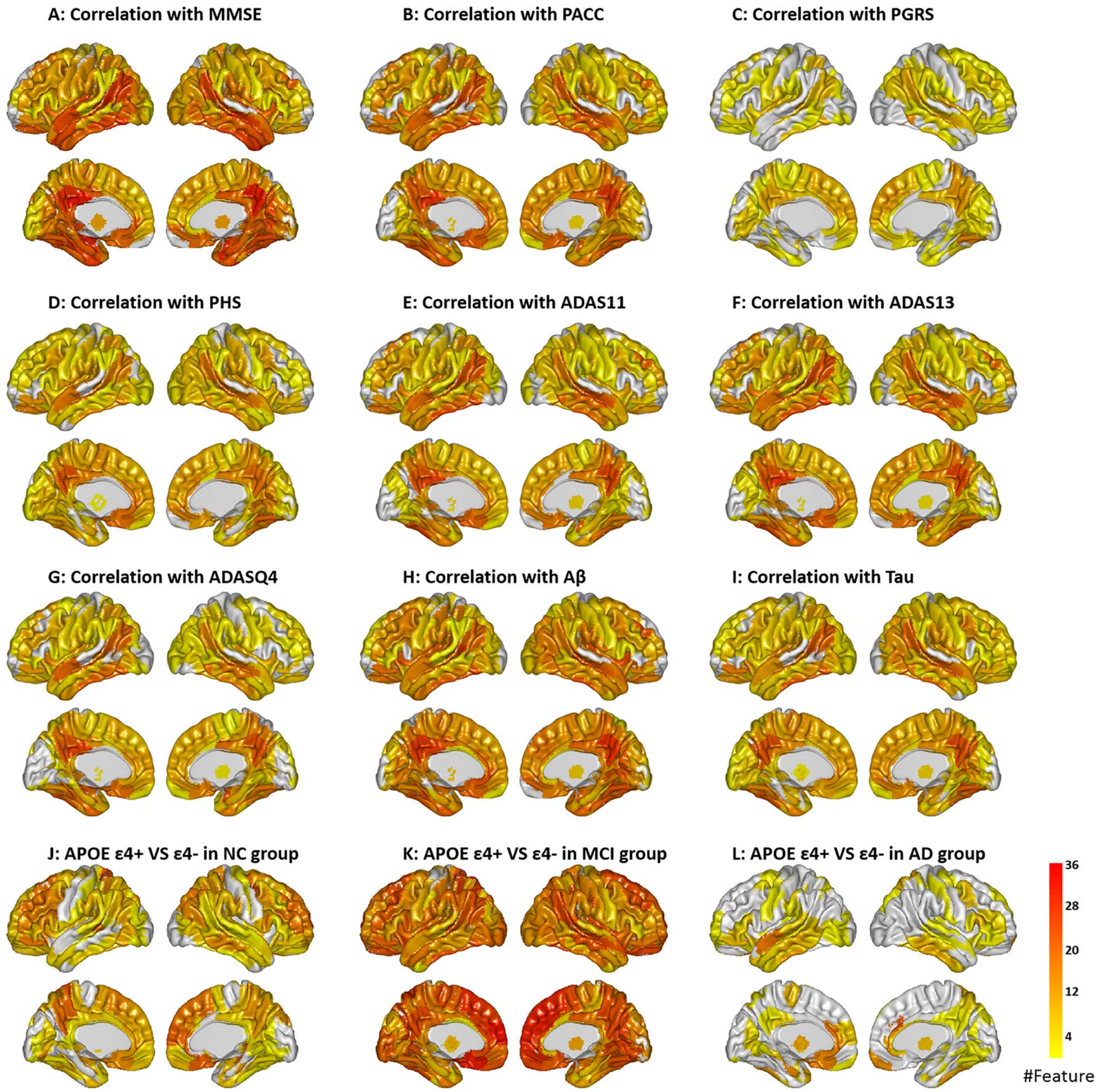

Figure 3. The results of the correlations between radiomics features and clinical measures including MMSE (A), PACC (B), ADASQ4 (C), ADAS11 (D), ADAS13 (E), PGRS (F), PHS (G), CSF Aβ (H), and CSF Tau (I). The altered radiomics features between APOE ε4+ and ε4− in the NC group (J), MCI group (K), and AD group (L). The value of the color bar indicates the number of features in each ROI in the Brainnetome Atlas.

SEN = 0.81, AUC = 0.90) (Fig. 4A, Table 2). For predicting whether MCI subjects will convert to AD with the baseline data, the ACC was 0.78 (SPE = 0.82, SEN = 0.74, AUC = 0.83) with LOOCV, and the average ACC was 0.72 (SPE = 0.76, SEN = 0.69, AUC = 0.75) with 10-fold cross-validation with the radiomics features of Aβ PET images (Fig. 4B, Table 2).

With the AD–NC classification model (NC is associated with a positive score, and AD is negative), we obtained the decision index of each MCI individual (the smaller the decision value, the easier it was to identify as AD). As shown in Fig 4C, the decision scores were significantly different between the subjects with or without APOE ε4+ in the MCI and AD groups (Supplementary materials S9), and we also found a significant difference in the decision value between the Aβ+&Tau+ and Aβ−&Tau− subgroups (Supplementary materials S5). Interestingly, the conversion time of PMCI significantly correlated with the decision value (P = 0.003) (Fig. 4D). In addition, the classification output significantly correlated with changes in cognitive ability (MMSE/ADAS13 at 12, 24, and 36 months) in MCI subjects (Fig. 4E–F, Supplementary materials S10).

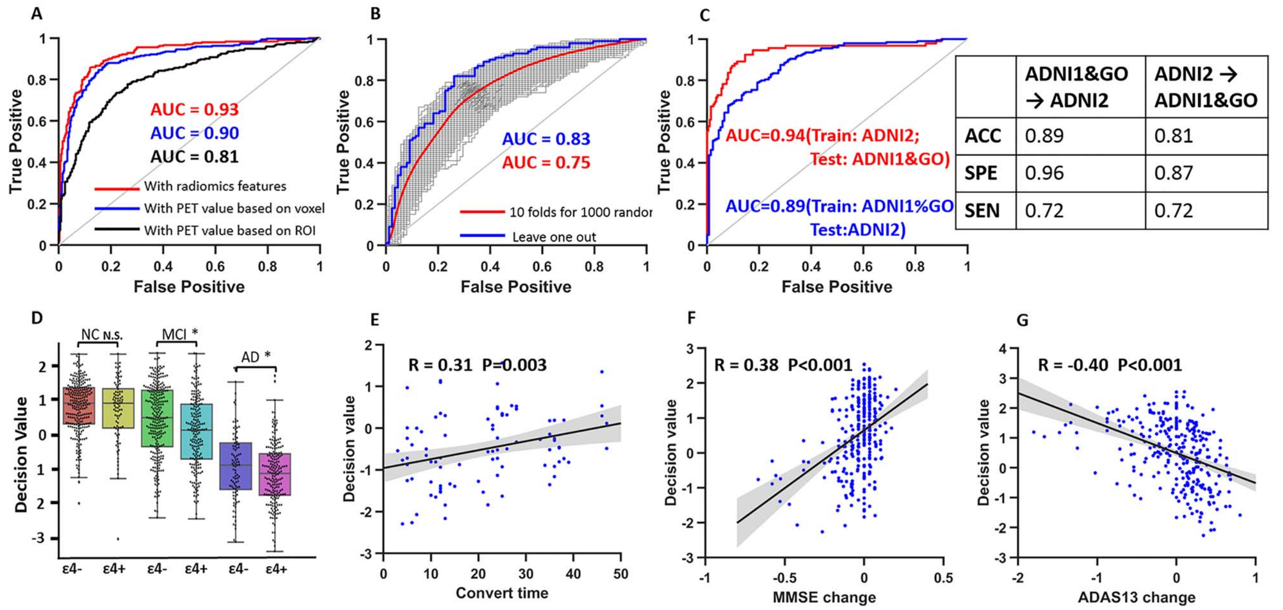

Figure 4. The ROC curves for the classification of AD and NC with radiomics features or PET values based on voxel or ROI (A); the classification of PMCI and SMCI via radiomics features with leave-one-out or 10-fold cross-validation (B); the ROC curves (ACC, SPE, and SEN) for the classification of AD and NC with ADNI1&GO as the training data and ADNI2 as the training data (C). The decision value with or without APOE $\varepsilon4+$ in the NC, MCI, and AD groups (D); the correlation between the time to conversion and the decision value in the PMCI subjects (E); and the correlation between the decision value and MMSE score (F), ADAS13 score (G) change in 36 months.

Table 2 The accuracy (ACC), sensitivity (SEN), specificity (SPE), and area under the curve (AUC) of the classification results

	ACC	SPE	SEN	AUC
CV1	0.87	0.88	0.86	0.93
CV2	0.75	0.80	0.70	0.81
CV3	0.84	0.86	0.81	0.90
CV4	0.78	0.82	0.74	0.83
CV5	0.72 ± 0.01	0.76 ± 0.02	0.69 ± 0.02	0.75

More importantly, we found that the classification outputs showed a significant correlation with MMSE scores in the MCI and AD groups ($R = 0.23$, $P < 0.001$) (Fig. 5A, Supplementary materials S11). In other words, the more severe the disease, the easier it was to be recognized by the classification model. In addition, the classification output significantly correlated with the other clinical measures (e.g., PACC, ADAS11, ADAS13, ADASQ4, PHS, CSF Aβ, and CSF Tau), and all P values were < 0.001 (and could pass through the Bonferroni correction) except for the correlation between output and PGRS (Fig. 5B–I and Supplementary materials S11).

Notably, the decision value of AD was computed with CV1 (10-fold cross-validation), which might be obtained with different model parameters. Thus, to remove the effect of decision value by the different models, we also computed the decision value with ADNI1&GO as the training data and ADNI2 as the testing data and vice versa. Significant correlations were also found between clinical information and decision value; details can be found in Supplementary materials S11.

The Associations between T-map of Difference Analysis and Gene Expression

The PLS method can estimate the correlation between the T-map of the identified radiomics feature (independent variable) and gene expression (dependent variables). The top two PLS components explained 45% of the variance in the radiomics response variables. The first partial least squares component (PLS1) and the second partial least squares component (PLS2) significantly correlated with the T-map of the altered radiomics features in AD, especially in certain identity features such as "Energy", "Mean", "Mad", "Median", "RMS", and some textural features such as "Cluster Prominence", "Cluster Tendency", "Homogeneity1", "RLN" and "RP" (Fig. 6 and Supplementary materials S12).

Discussion

In this study, we systematically demonstrated that the radiomic features of Aβ PET can serve as new neuroimaging biomarkers for clinical applications in AD with widely used machine learning and cross-validation techniques. Moreover, the comprehensive analyses of the associations between the features, classification output and clinical measures highlighted the solid biological/clinical basis underlying the progression of AD. This finding is of great significance for the early clinical diagnosis or prediction of AD and MCI.

Disease Severity-Associated Changes in MCI and AD

AD patients had reduced gray matter volumes and extracellular Aβ deposition in the medial temporal lobe, posterior cingulate

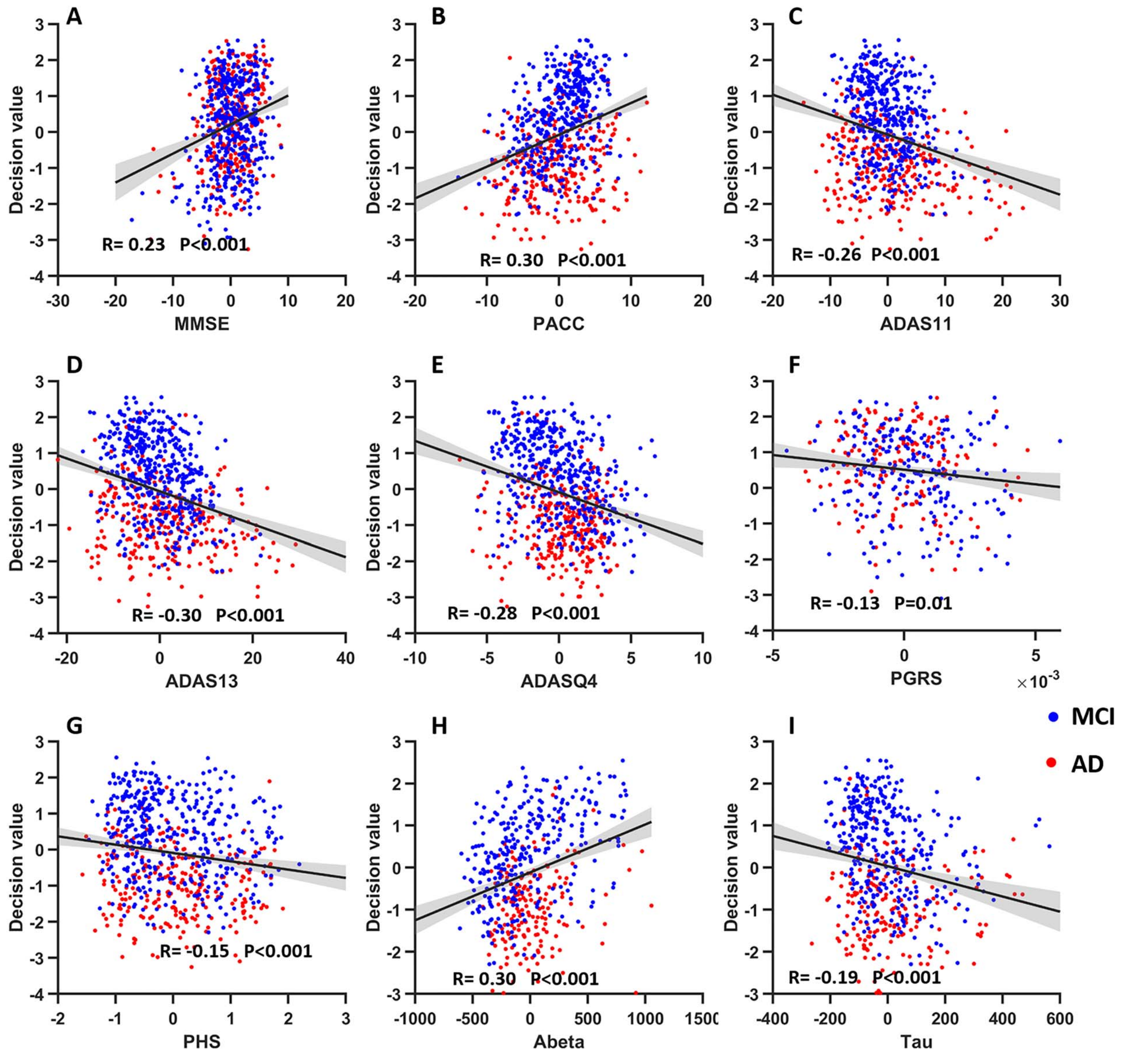

Figure 5. The correlation results between the decision value of the classifier and the MMSE score (A), PACC (B), ADAS13 (D), ADAS11 (C), ADASQ4 (E), PHS (F), PGRS (G), CSF Aβ (H), and CSF Tau (I). Note: The value of the clinical measures was plotted after regressing the age, sex, and group effects. The decision scores of the MCI subject were obtained based on the models of the classification of AD and NC. The decision value of AD was computed with CV1 (10-fold cross-validation), which was obtained with different model parameters.

gyrus, and superior frontal gyrus (Yang et al. 2012; Wang et al. 2015; La Joie et al. 2020). PET offers solid evidence for an increase in the accuracy of AD diagnosis (Jack et al. 2018). Our results showed the radiomics features of AD-related alterations in the temporal lobe, parietal lobe, and hippocampus and provided additional evidence of the Aβ burden that occurred in those specific brain regions.

Neuroimaging genetics aims to find the relationship between imaging markers and genetic phenotypes. As expected, the radiomics features showed significant differences between APOE $\varepsilon4+$ and $\varepsilon4-$ subjects in the patient groups, particularly in the MCI subjects. Notably, unlike a single gene locus, PHS and PGRS have been shown to have strong predictive ability and statistical power for evaluating the risk factors for AD (Desikan et al. 2017; Tan et al. 2018; Leonenko et al. 2019; Tan et al. 2019). The individual PGRSs that capture part of the degree of the risk of developing diseases and the PHS of AD have been proven to be enrichment markers for AD-associated Aβ and Tau deposition (Tan et al. 2019). In addition, the PHS and PGRS of AD are associated with cognitive decline and impaired brain imaging measures, highlighting the fact that elevated genetic risk influences traits even among individuals without dementia (Axelrud et al. 2018; Li et al. 2018; Logue et al. 2019). Hence, the associations between the PGRS/PHS and radiomic features highlight the genetic basis for Aβ PET in AD.

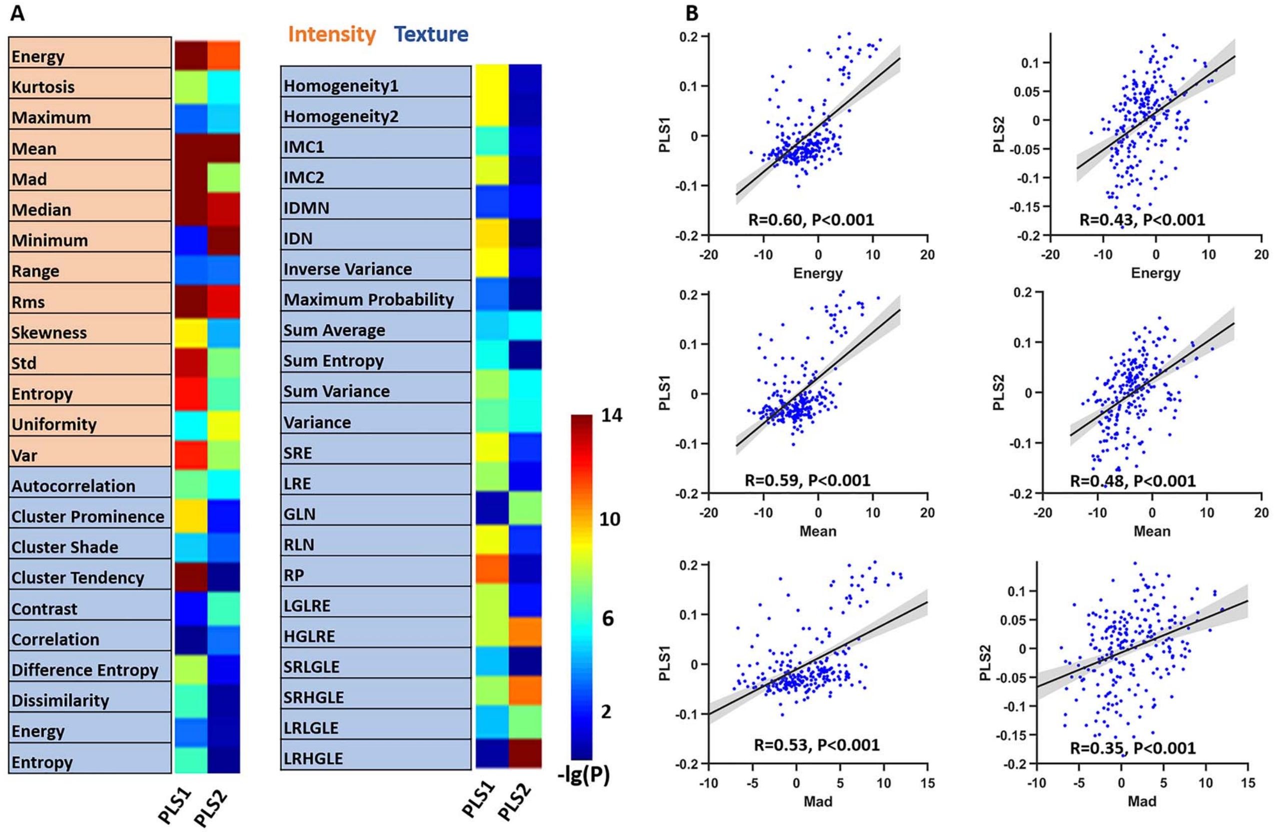

Figure 6. The results of the associations between T-map of difference analysis and the gene expression with PLS. The –lg(P) value of the correlation between the radiomics features and PLS1/PLS2 (A). An example (all belong to Intensity features) of the correlation between several radiomics features and PLS1/PLS2 (B).

Neurobiological Basis of Radiomic Features and Classification Output

Imaging biomarkers are the cornerstone of modern radiology, and the development of valid biomarkers is crucial for optimizing individualized prediction in AD (Dou et al. 2020; Jin et al. 2020a; Meyer et al. 2020; Zhao et al. 2020). The classification accuracy was similar to or higher than that in recent studies that were based only on the traditional classification model (Ritter et al. 2015; Beheshti et al. 2016; Schouten et al. 2016). Moreover, to the point of high accuracy, the mechanisms inherent in the models are particularly important for clinical applications. Identifying the biological basis of the imaging measures and classification output is clinically important for understanding the impaired brain architecture in AD (Jin et al. 2020b; Zhao et al. 2020).

The significant correlations between the imaging indices, classification outputs, and clinical scores confirmed that brain imaging alterations are associated with a solid neurobiological basis in AD and MCI subjects. More specifically, PHS was found to be associated with Aβ deposition and neurodegeneration in brain regions such as the rostral middle frontal cortex and the entorhinal cortex (Tan et al. 2018; Tan et al. 2019). The additional evidence that the strong correlations between the classification output and the PGRS, PHS, Aβ, and Tau implies that the higher the risk of the disease, the easier it is to recognize by the classification model, which illustrates a potential bridge between brain imaging and personalized medicine in AD (Lambin et al. 2017). We also found significant differences in the classification output between subjects with or without APOE $\varepsilon4+$ in the MCI and AD groups; hereby, the highly consistent results confirmed the genetic significance of the radiomics features of PET amyloid. A similar change pattern was identified in subjects with lower or higher risk as measured by CSF Aβ and Tau. Overall, these comprehensive pieces of evidence confirmed that radiomics features are potential imaging markers for AD translation. It is still a significant challenge to evaluate and predict the progression of MCI (Ding et al. 2019; Li et al. 2019a). The high performance of predicting whether MCI will progress to AD provides powerful evidence for using radiomics features to predict the progression of high-risk groups, which has important clinical implications. The association between the decision score and the duration of the converted time in the progression of MCI further indicated the possibility for clinical translation of the radiomics features (Li et al. 2019a, 2019b, 2019c; Zhao et al. 2020).

Caveats, Limitations, and Future Directions

Our findings should be interpreted with some additional considerations. AD is characterized by significant clinical heterogeneity, which is a key confounder for deepening our understanding as well as for enabling a more accurate diagnosis, prognosis, and targeted treatment (Ferreira et al. 2020; Habes et al. 2020). Creating a highly integrated novel framework for subtype analysis based on a large sample size, multisite and multimodal data offers additional evidence that holds great benefit for improving the individualized, precise, and early diagnosis of AD (Young et al. 2020; Qu et al. 2021). Imaging radiomic biomarkers offer an opportunity to improve patient diagnosis and to study the development of AD/MCI progression by clarifying the relationships between neurodegeneration and cognitive impairment. Deep learning neural networks (such as CNN and ResNet) have shown great potential in the classification of various diseases (Esteva et al. 2019; Li et al. 2019a; Qiu et al. 2020; Koppe et al. 2021). With the increase in data modalities from different sites, more advanced machine learning methods against radiomics features used in this work might help elucidate the neuropathological processes of AD. In addition, it would be very interesting to explore the robustness of the radiomics model in a population of test subjects that includes patients with other brain diseases, such as multiple sclerosis or Parkinson's disease. We explored the performance of the radiomics features of PET Aβ imaging only in the ADNI dataset. An independent dataset to verify the robustness of the results should be performed in future studies.

Supplementary Material

Supplementary material can be found at *Cerebral Cortex* online.

Notes

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Conflict of interest: The authors declare that they have no competing financial interests.

Authors' Contribution

Y.D. and K.Z. analyzed the data and performed the measurements; Y.D., K.Z., and Y.L. were principally responsible for preparing the manuscript; T.C., K.D., H.S., S.L., Y.Z., S.Y.L, and B.L. revised the manuscript; Y.L. supervised the project.

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